

ctrlX OS

Operating System for ctrlX CORE Control Devices
Version 4.4.1

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DC-AE/PAX (TaDo/PiaSt)

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1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2025-11	ctrlX CORE Runtime Version 4.4.1

Further information

- See [↗ ctrlX OS, ↗ Operating System for ctrlX CORE Control Devices, Application Manual](#)
- See [↗ ctrlX DRIVE, Firmware Release Notes AXS-V-06](#)

2 Important information

2.1 Information about our products

For the latest information on our products, go to [↗ https://www.boschrexroth.com](https://www.boschrexroth.com).

For current security advisories, please refer to [↗ https://psirt.bosch.com/security-advisories/](https://psirt.bosch.com/security-advisories/)

2.2 Information on the image update

The system images can be downloaded as zip archive from the Rexroth Collaboration Room.

NOTICE

As system images are device-specific, select the zip archive compatible with your device.

When using a system image, incompatible with your device, the device can sporadically not be used anymore, i.e. the device does not boot anymore.

In this case, please contact Bosch Rexroth support.

Further information on performing the image update can be found in the documentation "ctrlX OS Operating System for ctrlX CORE Control Devices", Application Manual in the chapter "[↗ Hardware reset for ctrlX CORE and ctrlX DRIVE Plus with integrated ctrlX CORE](#)".

2.3 LED display after an image update on a ctrlX DRIVEplus device can be incorrect



During an image update, the successful update might not be displayed correctly on a ctrlX DRIVEplus device.

For example that the LED is permanently blue or flashing red after a few seconds. At this point in time, the update is still running.

Thus, wait for 10 minutes after the start of the update.

After the waiting time, the device can be switched off, the micro SD card can be removed and the device can be switched on again.

Go to the log file on the SD card to check whether the update was successful.

3 Version 4.4.1

3.1 Software



This release is part of ctrlX CORE 4.4.0.

3.1.1 Components

Table 1: Included components

Short name	Software	Version	Additional notes
ctrlX Apps			
XCR-4.4.1	arch01.firstboot	4.4.1	
	arch02.uc24	4.4.1	
	arch02.engineering.uc24.firstboot	4.4.1	
	Hardware Support X3-4.4.0 (arch01.hw)	4.4.0	
	Linux Kernel X3-4.4.0+6.12.40-ctrlx-52 (arch01.kernel)	6.12.40-ctrlx-52	
	Hardware Support X7-4.4.0 (arch02-hw)	4.4.0	
	Linux Kernel X7-4.4.0+6.12.40-ctrlx-52 (arch02.kernel)	6.12.40-ctrlx-52	
	Hardware support OS-4.4.0 (arch03-hw)	4.4.0	
	Linux Kernel OS-4.4.0+6.12.40-ctrlx-52 (arch03.kernel)	6.12.40-ctrlx-52	
	Automation Core-4.4.5 (automationcore)	4.4.5	● ↗ New functions ● ↗ Bugfixes and modifications ● ↗ Note on use and known restrictions
	Device Admin-4.4.0 (deviceadmin)	4.4.0	● ↗ New functions ● ↗ Bugfixes and modifications ● ↗ Note on use and known restrictions
	Version Guard-4.4.0 (gating)	4.4.0	
	Setup-4.4.0 (setup)	4.4.0	● ↗ New functions ● ↗ Bugfixes and modifications ● ↗ Note on use and known restrictions
	Solutions-4.4.0 (solutions)	4.4.0	● ↗ New functions ● ↗ Bugfixes and modifications ● ↗ Note on use and known restrictions
	bare-1.0 (bare)	1.0	
	core22-20250612 (core22)	20250612	
	core24-20250729 (core24)	20250729	
	snappd-2.71 (snappd)	2.71	↗ https://github.com/snapcore/snappd/blob/master/packaging/ubuntu-16.04/changelog
XOH-4.4.0	XOH	4.4.1	
XOV-4.4.0	XOV	4.4.1+balanced	

3.1.2 Compatibility with drive-integrated controls

In the case of a drive-integrated control, ensure that the version of the control matches the version of the drive firmware.

The following table shows the possible combinations:

XCR control	Drive firmware
1.12	AXS-V-0306
1.12.15 and higher	AXS-V-0308
1.14	AXS-V-0308
1.16	AXS-V-0312
1.18	AXS-V-0314
1.20	AXS-V-0316 and AXS-V-0402
2.2	AXS-V-0404
2.4	AXS-V-0502
2.6	AXS-V-0504
≥ 2.6.10	≥ AXS-0504.4 ≥ AXS-0606.0
3.2	AXS-V-0602
3.4	AXS-V-0604
3.6	≥ AXS-0504.4 ≥ AXS-0606.0
4.2	AXS-V-0608
4.4	AXS-V-0610

3.2 New functions

3.2.1 Automation Core

Feature 740428: Improved handling of Motion errors

Description

Clicking on the "Reset errors" button in the axis commissioning dialog resets the associated Motion object in the Motion application and all associated errors and warnings in the alarms dialog. The same applies when deleting an associated Motion error in the alarms dialog. Previously, resetting the Motion object and deleting the error in the alarms dialog had to be carried out as two separate actions.

Feature 870976: Functional extensions in the logbook web interface

Description

The user rights for the logbook have been revised and adapted to the general user rights of ctrlX. While the old user rights regulated the availability of log message types (diagnostics, trace and system), the new user rights focus on the availability of functionalities ("view", "operate" and "manage"). After updating the software, the old rights continue to exist, but are marked as "deprecated". As soon as a new user right is assigned, the old rights lose their validity.

The handling of Motion errors has been improved.

The display of the columns in the default display of the logbook can now also be configured.

After an error reset, a message is displayed confirming that the reset has been completed.

There is a bookmark filter in the history overview that can be used to display only those entries for which a bookmark has been set.

The bookmark icon has been edited to improve the visualization of the difference between "set" and "not set".

All filter dialogs have been revised and standardized.

The date in the date column is displayed in the format that corresponds to the language set.

When the date filter is opened, the date of the currently selected entry is automatically entered in the filter.

Feature 938847: Updating the Celix Framework

Description

The Celix Framework has been updated to version 2.4.

Feature 984227: Updating the "core" system app

Description

The "core" system app has been updated to core24. This version is based on the Ubuntu 24.04 release.

Feature 995017: Various functional adjustments to the Data Layer web interface

Description

- Displaying the histogram data in a new area of the detailed view
For all nodes that have the "readType" "histogram", a new additional area is displayed in the detail view in which the data from the node is displayed in a chart diagram.
- Creating "raw" data is now possible
For nodes that have the "createType" "raw", creation with any files is possible.
- Creating "raw" data is now possible
For nodes that have the "createType" "raw", creation with any files is possible.
- Escape characters in the node names and addresses are taken into account
Addresses and names of nodes can now contain an escape character ('\\'). These are filtered out by the interface for display.
- Nodes with "eventType" can now also be subscribed to the Data Layer for reading the values
- Time stamps of the data can also be displayed in the input field
The display of the time stamp for the data can be enabled via a setting in the settings dialog.
- Optional scalars in JSON schema are now supported

Feature 996130: Changes in the Data Layer

Description

New functionality:

- Added environment variable DL_DEBUG to enable the debug output
- Task::onMetadata() added to change the meta data of a program task
- Version for 32-bit Linux
- InheritMutex and InheritRecursiveMutex are obsolete, use common::aos variants instead
- Events and diagnostics for connecting/disconnecting in devices/remotes
- Security enabled when used as a bundle

- Security can be enabled by defining certificate paths
- Escaping for addresses with ``
- ProgramTask now has the fields createdAt and finishedAt
- New flatbuffer version used (oss.flatbuffers 3.0.0+25.2.10)
- The static Data Layer library now contains all the required static libraries
- Remote Procedure Call (RPC)
- Removed dependency of trace and systemd libraries from comm.datalayer libraries
- Automatic provision of sub-nodes
- Websocket support
- Reverse connections are now enabled by default

Bugfixes:

- tokencheck.enable in celix config.properties did not work
- address2.h: escape() and unescape() are now publicly static
- RPC crash if no shared memory can be created
- SubnodeProvider:
 - onWrite did not work with asynchronous providers
 - Incorrect metadata for leaf nodes (cannot be browsed)
- Removing nodes of the subscribed list before calling onUnsubscribe the subscription nodes
- Static library now contains all required libraries (except openssl for security reasons)
- Thread name for identity change
- Possible crash if areas are registered via mddb
- Allowing the creation of areas for Asterisk nodes
- Possible crash if 8-bit ANSI is used as username/password in the connection string
- Improving the memory utilization of the Compliance Checker
- Possible crash when calling Variant::deserializeShared()
- Collision of shared memory areas in Windows
- openMemory() did not wait for the first connection attempt to the RT memory by default
- Possible crash if a variant with invalid string termination is received
- No subscription data on the remote node (Bug 835720)

Feature 1045881: Data Layer improvements

Description

New functionality:

- Loss-free subscription of entire rt/data nodes
- Nodes of RT link to master RT nodes in rtNode reference
- Writing shared retain data with Data Layer nodes
- findMDDb in C-interface
- The affinity of threads (providers and clients) can be configured in *framework/affinity/nrt-communication*
- Add *datalayer/statistics/provider/<id>/mddb* to display the provider's MDDBs
- Supporting the registration of subnodeprovider with bfbs instead of type strings

Bugfixes:

- No automatic reconnection of TCP client and providers when the broker is restarted
- Multiline description in the subnode provider
- Always inform if there is no connection established when waiting in `provider::start()`
- Failed start/stop of the provider when using thousands of nodes
- NVRAM synchronization counter only increases when NVRAM is actually updated

Feature 1047945: Flexible core division

Description

Below the Data Layer node *framework/affinity*, the core can be divided for the following areas:

- NRT communication
- Scheduler
- PLC Runtime

Compared to previous versions, threads can now use all cores for communication in the standard configuration and no longer just the first RT core. This means that communication with the control system is still possible in many applications despite the high system load.

Feature 1048023: Linux Kernel updated to version 6.12.40

Description

Update of the following components:

- Linux Kernel 6.12.40
- HailoRT v4.23.0
- NVIDIA Linux Open GPU Kernel Module 580.65.06

Feature 1059916: Functional extensions in the Data Layer web interface

Description

Several extensions have been implemented in the Data Layer web interface:

The "Data Layer node" widget is available on the start page, which can be used to display a value from a Data Layer node. Possible nodes that can be displayed in the widget have to be of type "number", "string" or "boolean". The widget display can be customized by the user depending on the node type.

Various dialogs in the Data Layer web interface (filter dialog for selecting big data nodes, dialog for creating child nodes) can be individually resized by the user.

All integer values (signed and unsigned) are displayed according to the set display format. The following formats (in addition to the decimal format) are possible: Bin, Oct and Hex. These integer values can also be edited in the writable input fields in all possible formats (decimal, bin, oct and hex). After editing, however, they are displayed again according to the set display format.

Online help can be called up for a node if the corresponding link is available in the meta data.

The node description is now displayed in multiple lines if the description contains a "\n" in the relevant places.

3.2.2 Device Admin

Feature 548599: Balanced energy mode for ctrlX OS (Balanced Mode)

Description

Two images are now provided for virtual devices. An image with which the virtual control is always operated at maximum energy utilization (performance mode). Furthermore, a second image is provided for virtual devices, which is operated with balanced energy utilization (balanced mode).

The switching function between the two modes is not yet implemented, but will be available in a future version of ctrlX OS.

Feature 836887: Device Admin supports CAN interfaces

Description

If a CAN interface is connected to a device, its baud rate can be set in the network settings and via a Data Layer command.

Feature 914768: Mounting and unmounting storage media using Data Layer commands

Description

Removable storage media can be mounted and unmounted via Data Layer commands.

Feature 948328: Simple Network Management Protocol can be used in ctrlX OS

Description

Simple Network Management Protocol (SNMP) is a network protocol that is used to monitor devices in a network. It enables the collection of information about the network device status.

Feature 986338: Notification for differing time

Description

If the ctrlX OS time differs from the browser time, this is displayed in the interface via a notification banner. The display continues across all pages until the time is correct again.

Feature 986818: Supporting VLAN via DAN network interfaces

Description

VLAN network interfaces with DAN network interfaces can be created as parent nodes.

Feature 996303: System report

Description

The system report contains information on the file system (Storage Media) and the RAM in the device-info.json file.

Feature 1006016: Renaming "Firmware update" to "Flash system image"

Description

The "Firmware update" tile under "Settings" has been renamed to "Flash system image".

Feature 1032629: Supporting Single Sign-on (SSO)

Description

It is now possible to log in to ctrlX OS using single sign-on via an identity provider. ctrlX OS has to be configured accordingly.

Feature 1040678: Improved display of the login banner for ctrlX OS

Description

When starting ctrlX OS, users are greeted with a terminal screen and a banner on the connected display.

All relevant network interface information is now displayed, which improves accessibility and user-friendliness.

- Displays all active IP addresses (IPv4 and IPv6) per interface.
- Simplifies the device connection from any network.
- Improves user-friendliness in virtual and multi-interface setups.

Feature 1054708: 10-day trial period available for selected apps

Description

Some apps, such as Firewall, Security Scanner and VPN Client, now support the one-time activation of a 10-day trial period directly under ctrlX OS.

Feature 1060885: Displaying the ctrlX OS version in "System information"

Description

The ctrlX OS version is set under *""Settings" → "System information" "* in the "General" section.

Feature 1061268: Display of remaining time in trial mode

Description

A ctrlX CORE virtual or an unlicensed installation of ctrlX OS switches off automatically after 4 hours. The time remaining until switch-off is displayed in the header.

Feature 1071645: Display CPU information in "System information"

Description

The CPU information is displayed under *""Settings" → "System information" "* in the "Hardware" section.

3.2.3 Setup

Feature 981999: Deleting backup files after restoring

Description

Backup files are deleted from the device after data has been restored successfully.

Feature 1048209: Supporting additional settings

Description

The following settings are also supported:

- CAN network settings
- Settings for the SNMP server
- Settings for Single Sign-On (SSO)
- Configuring network drives (NFS, SMB)

3.2.4 Solutions

Feature 995857: Configurations can be added to version control with Git

Description

Configurations can be version managed with Git.

The following functions are available:

- Adding archived configuration to Git
- Displaying and discarding local changes
- Tracking new and changed files
- Creating, checking out and comparing commits
- Show Git commit history
- Loading and saving the configuration taking into account the Git history
- Downloading, uploading and copying the configuration taking the Git history into account
- Managing Git settings on the device

Feature 995988: Collaborative engineering with Git

Description

Users can publish their changes in Git remote repositories. Branches can be used to manage commits and work together on technical tasks.

The following functions are available:

- Set up SSH connection
- Clone remote repositories
- Retrieving and transferring changes
- Creating and deleting branches
- Check out branches
- Merge branches
- Keeping an overview with the branch diagram

3.3 Resolved defects and modifications

3.3.1 Automation Core

Bug 1010221: Error when creating the callable during start-up

Description

In the PLC, you can use external events that automatically generate a callable in the ctrlX CORE Scheduler. During start-up, an error sporadically occurred when creating these callables.

Bug 1117971: Error when automatically creating callables

Description

When automatically creating callables, the error "Error during automatic reallocation of the callables" could occur.

3.3.2 Device Admin

Bug 940111: SD card can sporadically not be read in Windows PC after firmware update

Description

If an SD card is partitioned via the partition management in the "Memory" interface, it is detected under Windows.

However, if the card is partitioned by means of a firmware update, the storage medium sporadically cannot be read under Windows.

Bugfixing

Before creating the new partition table, all existing partitions are first deleted. The SD card is subsequently also identified in the Windows PC.

Bug 994012: "Upgrade ctrlX OS" (remodeling) fails sporadically**Description**

If "Upgrade ctrlX OS" is selected on the apps page, the error message "Remodeling execution failed" is reported sporadically.

Bugfixing

"Upgrade ctrlX OS" (remodeling) is repeated if it fails.

Bug 995534: A USB flash drive cannot be mounted again if it is removed without clicking on "remove safely"**Description**

If you remove a mounted storage medium without clicking on "remove safely", the storage medium cannot be mounted again without restarting the device.

Bugfixing

If you remove a mounted storage medium without clicking on "remove safely", the storage medium can be re-inserted and mounted again.

Bug 1043154: When formatting a partition, the name of the storage medium must be specified**Description**

According to the API, formatting a device using the "media" or "device" parameter is possible. Due to another bugfix, formatting was currently only possible with the "device" parameter.

Bugfixing

Formatting is now possible again, as specified, using the "media" or "device" parameters.

Bug 1049110: Snapd app update fails for devices without serial assertion**Description**

The serial assertion is usually created automatically when the device is connected to the internet and authenticates itself with the ctrlX Store. If a device has not received a serial assertion since the last image update, an error occurs when updating the Snapd app.

Troubleshooting

Since then, the installation process has only waited a certain amount of time until the device is ready for installation. It is now ensured that the system waits until this state is reached.

Bug 1064879: Alarm "Failed to set CPU configuration" is displayed**Description**

The alarm "Failed to set CPU configuration, preconditions not met" is incorrectly displayed as an error in the logbook. The reason for this is the change-over from cgroups-v1 to cgroups-v2.

Bug 1079925: "Upgrading ctrlX OS" (remodeling) not possible with snapd 2.68.5.1

Description

If ctrlX version 1.20 has been installed on a device and all system apps have been updated to version 3.6 (including snapd 2.68.5.1), "Upgrade ctrlX OS" is displayed on the "Apps" page. If you click on this, the error message "Remodeling execution failed" is displayed.

Bugfixing

Snapd 2.71 supports "Upgrade ctrlX OS" (remodeling) again.

Bug 1081521: The system operating time is not updated in the Data Layer

Description

The system operating time is only updated once when the control unit is started in the Data Layer.

Bugfixing

The system operating time is continuously updated in the Data Layer.

Bug 1126921: Error when restoring a subscription (certificate management)

Description

If an attempt is made to change an existing subscription as part of a restore operation for an existing subscription, this leads to an error 404 ("Subscription id not found").

Bugfixing

Subscriptions can be restored.

3.3.3 Setup

Bug 1015711: Resetting the default password is not possible

Description

If a default password was set in the setup app, it could not be reset.

Bug 1035162: Unauthorized access to sensitive data via security mechanisms

Description

A vulnerability in the web application of the ctrlX OS setup mechanism facilitated an authenticated (low privileged) attacker to gain remote access to backup archives created by a user with elevated permissions. Depending on the content of the backup archive, the attacker may have been able to access sensitive data.

Troubleshooting

Downloading the backup file is only permitted for administrators with "Full access" authorization.

Bug 1035172: Unauthenticated access to API endpoint

Description

A vulnerability in the Task API endpoint of the ctrlX OS setup mechanism allowed a remote, unauthenticated attacker to access and extract internal application data, including potential debug logs and the version of installed apps.

Troubleshooting

The Task API provides only very limited information to unauthenticated users. Only users with "Full access" receive further information.

Bug 1041006: Backup/restore does not support NFS and SMB mount points

Description

When selecting an NFS or SMB mount point to create or restore a backup, an error occurred and the process terminated with an error message.

Bug 1053299: Unencrypted backup files

Description

Ambiguous wording in the web interface of the ctrlX OS setup mechanism could lead the user to believe that the backup file is encrypted when a password is set. However, only the private key code - if available in the backup - is encrypted, while the backup file itself remains unencrypted.

Troubleshooting

"Encrypted ZIP file" is selected as the default format for backup files and the outdated description above the password field has been removed.

Bug 1062953: Certain snapd versions cannot be restored

Description

Certain snapd versions were not detected correctly when restoring a backup. As a result, snapd may be sporadically not updated and subsequent errors occurred, causing the recovery to fail.

Bug 1068029: Error when restoring PKIs

Description

PKIs that use a certificate that is not stored in the "Network Security" certificate store, but in a different one, such as "Data Layer" could not be restored.

Bug 1071563: User-defined password policy is not deleted when restoring a backup

Description

If a user ("user") was assigned a password policy ("named password policy") added by the user, and the user and the password policy were subsequently deleted by applying a backup, the error was reported that the password policy is in use and therefore cannot be deleted.

Bug 1076429: User is not removed from group

Description

If a user is a member of at least one group, but not part of a group in a backup file, and the backup is restored in "Restore" mode, the user is not removed from the group(s).

Bug 1077634: Server-based licensing is not enabled when restoring a backup

Description

Server-based licensing could be disabled by applying a backup, but not enabled.

Bug 1077648: Warning "Json schema issue: ..." when restoring a backup**Description**

If an individual certificate and key are selected for client authentication ("individual certificate and key") in a backup with server-based licensing, two warnings, each starting with "Json schema issue: ...", are incorrectly issued when the backup is restored.

Bug 1079332: License server is not enabled when restoring a backup**Description**

If the license server functionality ("floating license server") was disabled on a ctrlX device, the functionality could not be enabled by restoring a backup.

Bug 1113522: Certificate and PKI cannot be deleted together**Description**

If a certificate is used in a PKI and both are not included in a backup, neither can be deleted by applying the backup in "Restore" mode.

3.3.4 Solutions**Bug 970580: Authorization "Manage configurations" is insufficient for loading and saving****Description**

Depending on the app installation, authorization can be missing for users with the "Manage configurations" authorization, which results in failed loading or saving (403 Forbidden).

Troubleshooting

The "Manage configurations" authorization allows the user to load and save configurations (independently of other installed apps).

Bug 1070473: Uploading archives with read-only directories fails**Description**

Configuration archives (ZIP files) could not be uploaded to a device if directories within the archive were read-only.

3.4 Notes on use and known restrictions**3.4.1 Updating the apps on the ctrlX CORE Virtual**

The setup ctrlX WORKS provides a new device ctrlX CORE Virtual compatible with the current version, including the current system apps. To convert an existing application to this version, proceed as follows:

Execute the following steps in the described sequence:

1. ➤ First connect to the ctrlX CORE Virtual and go to menu item "Manage app data" via  in the header or via "Home" → *tile "Manage app data"*.
2. ➤ Back up the current app data via  "Save to archive".
3. ➤ Save all available configurations by highlighting the respective configuration in the list and click on the button .
4. ➤ Create a new ctrlX CORE Virtual in ctrlX WORKS in the "Devices" window.

5. ➤ Install the application apps in ctrlX CORE.

 The application apps can be installed in the “Operating” state as long as not real-time apps (Motion, EtherCAT, PLC, PROFINET Device) are installed.

When a real-time app is installed or available on the system, switch the “Operating” state to “Service” before the installation.

- First install all application apps via the button  “Install from file”.
- Highlight all application apps and confirm them in the following file drop-down box. Subsequently, these apps are bundled and loaded to the control via “Install”.
- Wait until loading is completed.
- Now press the button “Update all apps”. The apps are installed.
- Switch to the state “Service”.
- Update all real-time apps (Motion, EtherCAT, PLC, PROFINET Device).
- Click on “Operating”.
- Subsequently, refresh the page to display the new system state.
The control is ready again for operation.

6. ➤ Go to “Manage app data” and upload the relevant configuration backups via the button. Subsequently, load the configuration to be executed via the button. The update is completed.

7. ➤ Please verify that all data has been applied.

Further information can be found in the chapter ➔ [Configurations – Introduction](#), ctrlX OS, Operating system for ctrlX CORE Control Devices, Application Manual.

3.4.2 Updating the apps on ctrlX CORE

Check the version of the installed apps on the ctrlX CORE control.



The following instructions are only valid if at least release version ctrlX CORE at least the release version XCR-V-0202 or a subsequent software version is installed.

Preparation: The apps to be updated were copied to a local directory on the PC on which the update can be executed.

The system apps can be installed in the “Operating” state as long as no real-time apps (Motion, EtherCAT, PLC, PROFINET Device) are installed.

As soon as a real-time app is installed or is available in the system, the status has to be changed from “Operating” to “Service” before installation.

Execute the described steps in the described sequence:

1. ➤ First connect to the ctrlX CORE and go to menu item “Manage app data” via  in the header or via “Home” → *tile “Manage app data”*.
2. ➤ Back up current app data via  “Save to archive”.
3. ➤ Save all available configurations by highlighting the respective configuration in the list and click on the button .
4. ➤ Select the menu item “Settings → Apps” to go to the overview of the installed apps.

5. Install the system apps now.

- Install all system apps via the button  “Install from file”.
However, if the apps are installed individually, the following sequence must be followed:
 1. Update the "snapd" system app.
 2. Install the "core24" system app.
 3. Update the "Device Admin" system app.
 4. You can then install or update all other apps using the “Install from file” button.
- In the following file selection window, select all the desired apps and confirm your selection. Subsequently, these apps are bundled and transmitted to the control.
- Wait until loading is completed.
- The update information for the apps already available on the control and the new apps to be installed are displayed in a notification window.
Press the button “Install” and confirm the dialog.
- Acknowledge the following messages with “Continue”. This starts the installation.
- After the installation, log into the ctrlX CORE again or go to the “Apps” window.
- Subsequently, refresh the page to display the new system state.
- The update of the system app has been completed.

6. Install the required application apps.

The next steps are always required, even if the system apps have been installed via the rescue system or you have received a control on which the current system apps are already installed. If the control is not yet in this state, first press the “Service”).

- Install all application apps and real-time apps (Motion, EtherCAT, PLC, PROFINET Device) via the button  “Install from file”.
- In the following file selection window, select all the desired apps and confirm your selection.
- Press the button “Install” and confirm the dialog.
- Acknowledge the following messages with “Continue”. This starts the installation.
- ➔ The control is ready for operation again.

7. Go to “Manage app data” and upload the relevant configuration backups via the button. Subsequently, load the configuration to be executed via the button.

- ➔ The update is completed.

3.4.3 Automation Core

Bug 1051857: X3: Pending error after image update**Description**

During the image update, the error "Condition detected that triggered a safe machine state" can occur.

Workaround

A reboot fixes this error.

Bug 1098779: System message "Cannot find unit for notify message of PID **, ignoring." in the logbook****Description**

If you activate the system messages in the logbook, there are many entries in the logbook with the text "Cannot find unit for notify message of PID ****, ignoring.". This does not affect the functionality of the ctrlX and is only a message from the system.

Bug 1100451: System image update from version 2.6 and older to version 4.4 via the ctrlX OS web interface results in an error**Description**

A system image update from version 2.6 and older to version 4.4 via the "Firmware Update" window in the ctrlX OS web interface results in an error and is therefore not possible.

Workaround

Download the ctrlX CORE system image from the collaboration room and create the USB flash drive for the firmware update using the "ctrlX OS Recovery" function in ctrlX WORKS.

Bug 1111432: ctrlX CORE X5/X7: Warning 080E082A/0C64100D "App connection restored" after image update**Description**

With the ctrlX CORE X5/X7, the warning 080E082A/0C64100D "App connection restored - Missing app connection of the signed application is being restored" is displayed in the logbook after the image update upon the first startup. The reason for this is that an app interface has to be connected manually. After updating the image, the initial start-up is delayed by 10 to 20 seconds.

Bug 1120682: ctrlX CORE X7 startup sporadically delayed by 140 seconds**Description**

The startup of a CORE X7 may be sporadically delayed by 140 seconds.

Bug 1122688: Increased runtime of scheduler tasks**Description**

The maximum runtime of the scheduler tasks has been increased by up to 20% in version 4.4.0 compared to version 3.6.3.

Bug 1126017: Frequent error 081F0042/0C7B4285 when switching from OPERATING to SERVICE**Description**

When switching from OPERATING to SERVICE, the error "081F0042 Error in Scheduler system / 0C7B4285 Switching state returned 'SCHED_E_LOCK_TIMEOUT'" can occur frequently. This error can be confirmed with PROCEED. Switching takes place correctly despite the error message.

Bug 1131271: Increasing "deviation" time measurement for scheduler tasks**Description**

The "Deviation" time measurement on a CORE X5/X7 results in an incremental increase in time.

3.4.4 Device Admin

Bug 855538: "Reset device data" can fail sporadically

Description

When executing the setup, an error occurs which results in an error message after completing the operation.

Bug 1021621: Hardware support cannot be updated on a control with Secure Boot (ctrlX CORE X5/X7) from the store

Description

Attempting to update the Hardware Support App from the store fails as the SBAT version, which is important in connection with Secure Boot, cannot be determined. The installation from the store is therefore rejected.

Bug 1076361: Time and time zone are not applied

Description

Ubuntu Core 24 no longer supports some time zones (e.g. Asia/Calcutta). However, these outdated time zones can still be selected in the web interface. If such a time zone is selected, the time information is not applied. However, no error message is displayed.

Workaround

Use a supported time zone.

Bug 1111609: Updating all apps on a device with image version 2.6.10 is canceled after the app Core22

Description

When trying to update all apps on a device with ctrlX OS image version 2.6.10 via batch upload, e.g. to version 3.6.3, the process is canceled after the Core22 app.

Workaround

First update the core app. All other apps can then be updated simultaneously.

Bug 1116146: ctrlX OS sporadically takes longer to boot on ctrlX CORE

Description

Updating the kernel from 6.1 to 6.12 can cause the internal eMMC controllers on a ctrlX CORE to be initialized in the "wrong" order. The internal memory is connected to one of the two controllers, the other controller is not connected. If the unconnected controller is initialized first, the name of the internal storage medium changes from /dev/mmcblk0 to /dev/mmcblk1. Ctrlx-vault is then persistently mounted on this device. Upon the next restart, snapd tries to mount the ctrlx-vault on this device. However, if the connected controller is faster during this startup, it is assigned the name /dev/mmcblk0 and the mounting freezes for 90 seconds before the controller completes startup.

Bug 1126501: After updating all system apps on devices with image 2.6, "Upgrade ctrlX OS" (remodeling) cannot be executed

Description

On devices with image 2.6, it is not possible to run "Upgrade ctrlX OS" (remodeling) after updating all system apps. It is canceled with an error message.

Workaround

The current image with version 4.4 has to be installed. The bug is planned to be fixed in the next version 4.6.

Bug 1129085: Image update does not work reliably on ctrlX CORE X5

Description

The internally connected eMMC controller is not reliably initialized as mmcblk0 on ctrlX CORE X5 with kernel 6.12 (included in image 4.4), but with a probability of 50% as mmcblk1.

For the image update, the file rescue.toml is written, which defines the target medium for the image.

If the mmcblk1 medium is now detected as the boot medium before the image update, this is also written to rescue.toml and the image update cannot be carried out.

Workaround

Before executing an image update, check which medium is detected as SystemData. If it is not mmcblk0, reboot the system prior to updating the image.

3.4.5 Setup

Bug 1116120: Backup/restore may fail after 30 days operating time of the device

Description

Backup/restore and requires an app-specific temporary folder. This folder is created upon startup. With CORE24, a change was introduced with regard to cleaning up temporary folders. With the current version used by ctrlX OS, temporary folders that are not used for 30 days are removed. After removing the temporary folders, an app can no longer access its app-specific temporary folder, which can result in various errors, e.g. backup/restore can no longer fulfill its task, apps can no longer be installed, etc.

Workaround

If the operating time of a device is more than 30 days and apps cannot perform their tasks, it may be helpful to restart the device. The desired task should then work again, provided the error was related to the problem mentioned.

3.4.6 Solutions

Bug 1117106: Files with non-ASCII characters in the name are restored incorrectly when uploading

Description

If a configuration contains files whose file names contain non-ASCII characters and this configuration is archived with a ZIP program that does not automatically use UTF8 for such characters (e.g. 7-Zip), the names of these files are restored incorrectly when uploading. The Windows-integrated packer and the Solutions app use UTF8 when creating archives, so that the problem does not occur.

Bug 1122488: Incomplete backup of Git settings

Description

The settings for the SSH remote server are not backed up and restored.

4 Service and support

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- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)

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